

Secure Systems Engineering

EE G7701

# VOTING MACHINE: USER MANUAL

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# PREFACE

This document explains how to use the voting machine.

The voting machine is a tool that allows us to perform elections. In that matter, it fulfills the following functional requirements:

## **Voter Registration**

- Eligible users register as voters by providing their full name and date of birth.

## **Admin Authentication**

- Admins are required to log in using a secure username and password before managing any aspect of the election process.

## **Ballot Creation**

- Admin can create election ballots. Each ballot has multiple offices. Each office displays candidates and their parties.

## **Opening and Closing Ballots**

- Admins can open or close ballots for voting.

## **Vote Casting**

- Registered voters can select a ballot and cast votes for candidates in different offices.

## **Vote Tallying**

- Once ballots are closed after an election, the admin can tally votes for all candidates.

## **Deletion**

- Apart from votes, all elements can be deleted by the admin. Offices, voters, and candidates can be deleted as requested by the system admin.

The voting machine is fully developed in Rust programming language. It stores all data in an integrated SQLite database. Tables, with strong key requirements, are created and populated as per user order.

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# 1 Downloading and Installing the Project

## STEPS

1. The installation process mentioned below will be the similar for all operating systems (Including installation inside a VM Machine).
2. Download the project from the given source.
3. Unzip the project and place it in your home directory or working environment.
4. The project is integrated using SQLite database to store the data.
5. To view the data in your database system, download the SQLite Database.
6. Visit the following link - <https://www.sqlite.org/download.html>
7. Download the correct file that supports your system OS.
8. Navigate to the project and open your chosen IDE/Terminal.

Please note that this part of the manual assumes that you already have installed all tools necessary to compile and run a code. When it comes to the voting machine, the software is auto-sufficient as the cargo.toml file contains all the dependencies to be downloaded when compiling the code.

# 2 Running the project

## STEPS

1. Open your terminal.
2. Navigate to the project directory using the “cd” command where you placed the project.
3. Once you are inside the project directory, run “Cargo build” to build the project first or you can use “Cargo run” to directly build and run the project.

After compilation, the terminal will show you a prompt for the voting machine looking like the image below:

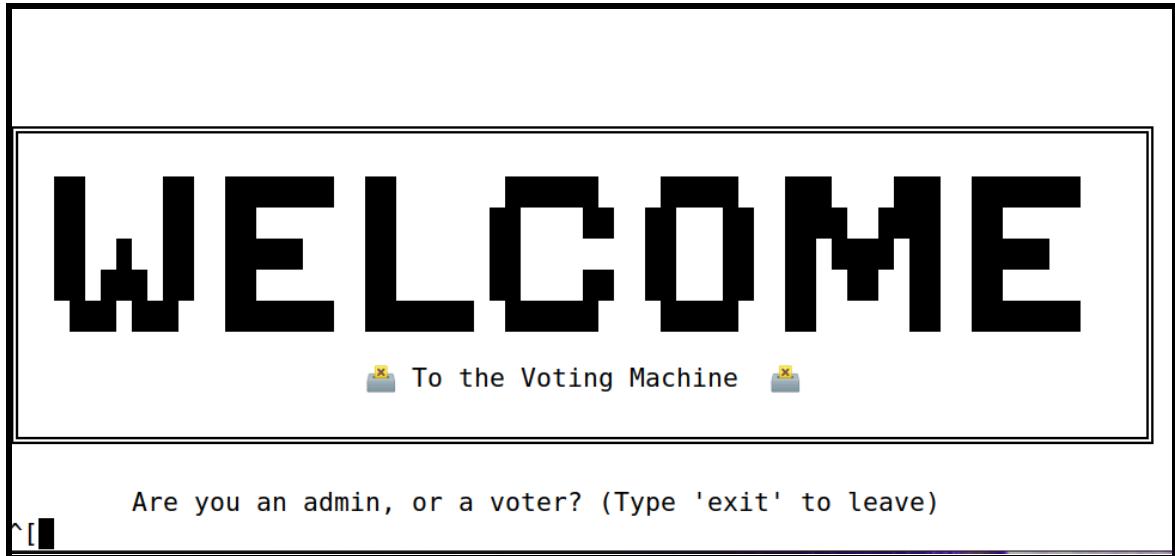


Figure 1: Landing page of voting machine

### 3 Log In as a System Admin

#### STEPS

1. Once prompted the question on the screen, “Are you an admin, or a voter?”; Type “admin” to use the system as an admin.
2. The system will prompt you with a screen asking for admin username.
3. Provide “**adminname**” as your username input.



Figure 2: Admin username insertion

4. Provide “**adminpassword**” as your password input.



Figure 3: Admin password Insertion

5. The system will show you an Admin Menu after successfully logging in as an Admin.

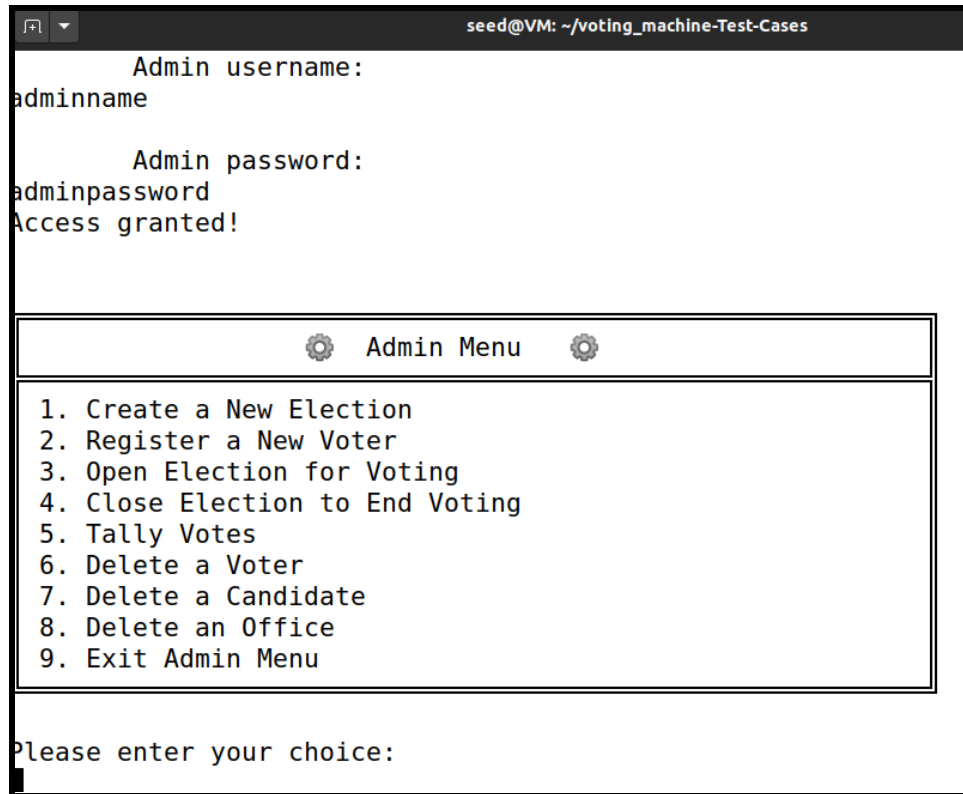


Figure 4: Admin menu display

6. According to the Menu enter your choice by providing the numbers such as “1,2,3 etc”.
7. Following steps will guide you for every selection starting with number 1.
8. Type “1” and press enter for “Create a New Election” option.
9. You will be prompted to enter the name of the office.



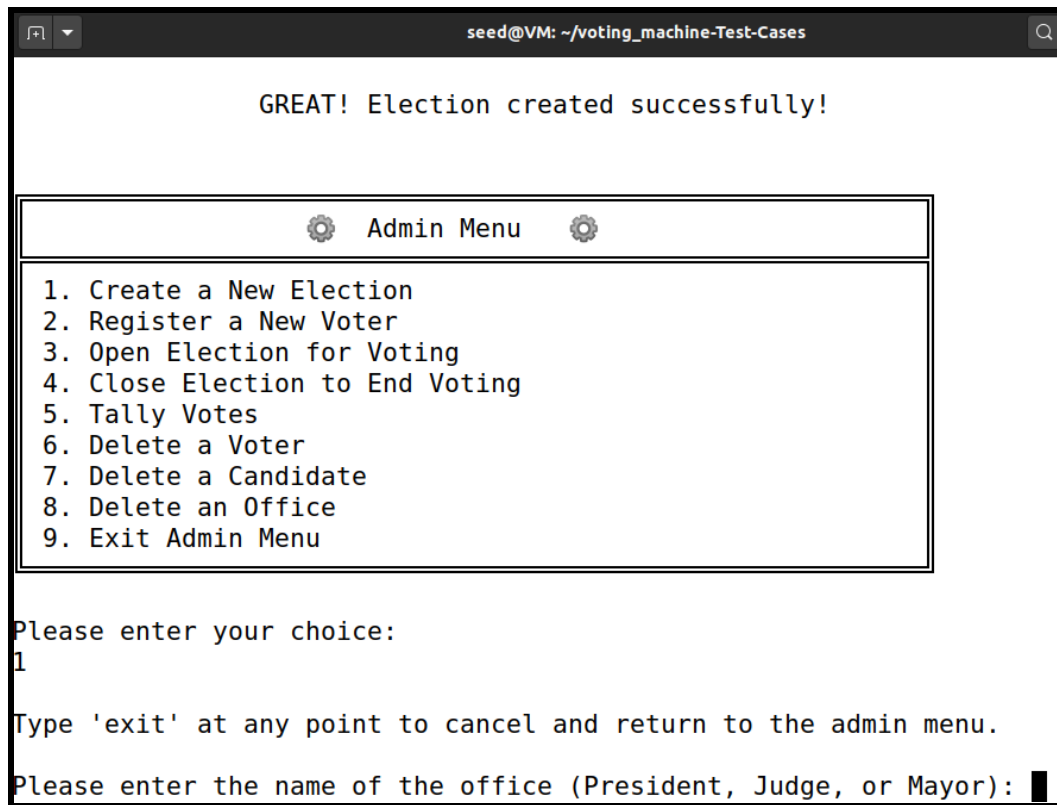


Figure 5: create a new election

10. Select your preference of the office (e.g, Judge).
11. You will be prompted to enter the name of the candidate.
12. You will be prompted to enter the political party of the candidate.
13. Once you enter your desired input. You will be asked if you are done adding or you want to add more candidates. If not, answer “no” and press enter.
14. The system will show you a message that your election is created.

```
Please enter the name of the office (President, Judge, or Mayor): President
An office with the name 'president' already exists!

Type 'exit' at any point to cancel and return to the admin menu.

Please enter the name of the office (President, Judge, or Mayor): Judge
Please enter the name of the candidate: Gallileo
Please enter the political party of the candidate: Marxist
Add another candidate to the office (type 'yes' or 'no')?: no
Add another office to the ballot (type 'yes' or 'no')?: no

                GREAT! Election created successfully!
```

Figure 6: Successful creation of Office

15. Type “2” and press enter for “Register a New Election” option.
16. The system will prompt you with a screen asking for voter’s information.

seed@VM: ~/voting\_machine-Test-Cases

Admin Menu

1. Create a New Election
2. Register a New Voter
3. Open Election for Voting
4. Close Election to End Voting
5. Tally Votes
6. Delete a Voter
7. Delete a Candidate
8. Delete an Office
9. Exit Admin Menu

Please enter your choice:  
2

VOTER REGISTRATION

Please provide the following information.  
Type 'exit' at any time to cancel.

Enter voter's complete name: Alex turner  
Enter voter's date of birth (MM/DD/YYYY): 01/06/1986

Figure 7: Register a new voter

17. Type “Yes/No” according to the correctness of the provided information.

```
seed@VM: ~/voting_machine-Test-Cases
7. Delete a Candidate
8. Delete an Office
9. Exit Admin Menu

Please enter your choice:
2

VOTER REGISTRATION

Please provide the following information.
Type 'exit' at any time to cancel.

Enter voter's complete name: Alex turner
Enter voter's date of birth (MM/DD/YYYY): 01/06/1986

Confirm the following details:

Name: alex turner
Date of Birth: 01/06/1986

Is this information correct? (yes/no):
```

Figure 8: Voter details verification

18. If the information is correct you will see a prompt saying that the voter registration is successful.
19. Type “3” and press enter for “Open Election for Voting” option.
20. The screen will show you a message saying “The election has been opened for voting.”
21. Type “4” and press enter for “Close Election to End Voting” option.
22. The screen will show you a message saying “The ballot has been closed. No voting possible!”

```
seed@VM: ~/voting_machine-Test-Cases

Please enter your choice:
3
The election has been opened for voting.

Admin Menu

1. Create a New Election
2. Register a New Voter
3. Open Election for Voting
4. Close Election to End Voting
5. Tally Votes
6. Delete a Voter
7. Delete a Candidate
8. Delete an Office
9. Exit Admin Menu

Please enter your choice:
4
The ballot has been closed. No voting possible!
```

Figure 9: Opening or closing an election

23. Type “5” and press enter for the “Tally Votes” option.

24. The screen will show you the result of the Elections.

```
seed@VM: ~/voting_machine-Test-Cases

Please enter your choice:
5
----- ELECTION RESULTS -----

Office: Judge
  Oumou (Tree) - 0 votes
Office: Mayor
  C1 (P1) - 0 votes
Office: Mayor
  c2 (p2) - 0 votes
Office: Officel
  mere (mire) - 0 votes
Office: President
  Alice (Party A) - 3 votes
Office: President
  Alice (Party A) - 2 votes
Office: President
  Alice (Party A) - 1 votes
Office: President
  Maimouna (Democrats) - 0 votes
Office: President
  Bryan (Liberal) - 0 votes
Office: President
```

Figure 10: Tally votes

25. Type “6” and press enter for the “Delete a Voter” option.

```
seed@VM: ~/voting_machine-Test-Cases
Office: exit
exit (9) - 0 votes
```

⚙ Admin Menu ⚙

1. Create a New Election
2. Register a New Voter
3. Open Election for Voting
4. Close Election to End Voting
5. Tally Votes
6. Delete a Voter
7. Delete a Candidate
8. Delete an Office
9. Exit Admin Menu

```
Please enter your choice:
6
Type 'exit' at any time to cancel and return to the admin menu.

Enter the name of the voter to delete:
```

Figure 11: Delete a voter

26. Type “7” and press enter for the “Delete a Candidate” option.







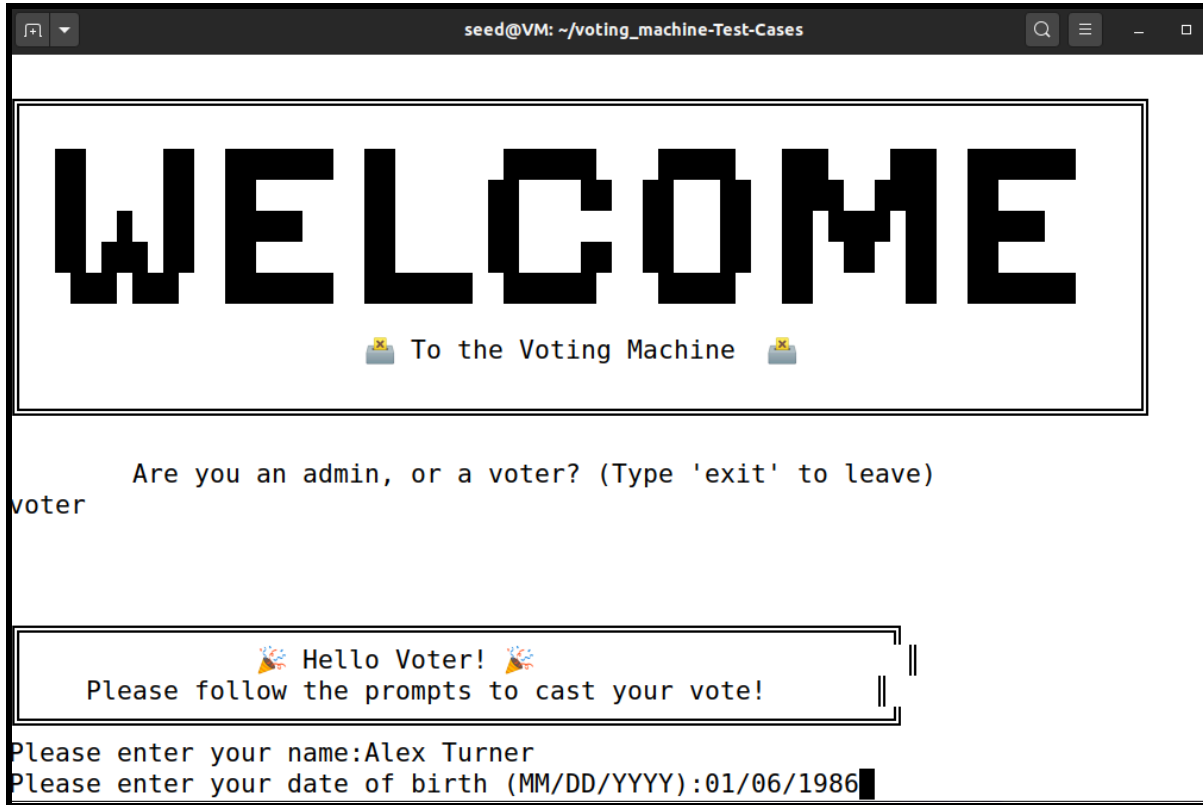


Figure 14: Log in as a voter

4. After verifying your information the system will welcome you to the system to provide your vote for the office.
5. Make sure to open the election using admin privilege. The process is described in the "For an Admin User" section. Remember if the election is closed the voter will not be able to vote.
6. After getting access the system will show you the available offices that are created by admin.

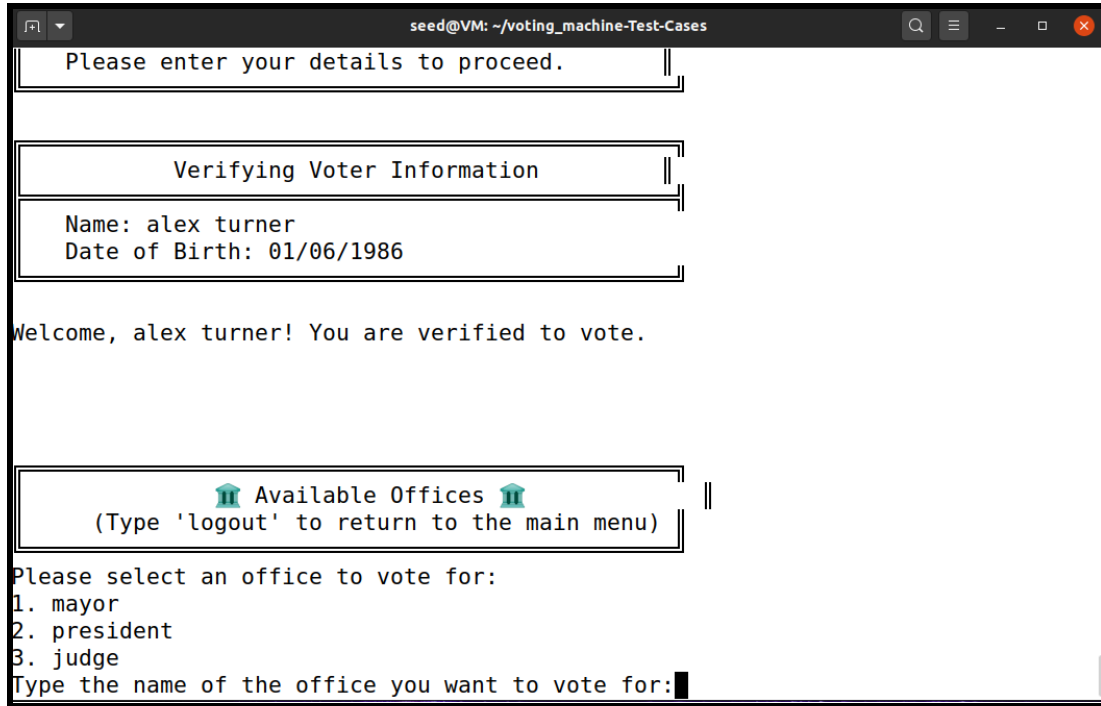


Figure 15: Voting for a candidate- selecting the office

7. Type the name of the office that you want to vote from the available options.
8. After selecting the office the system will show you the candidates for that office.
9. Provide the name of the available candidate in the input field, once prompted.
10. Press enter and you will be shown a successful message that your vote has been casted.

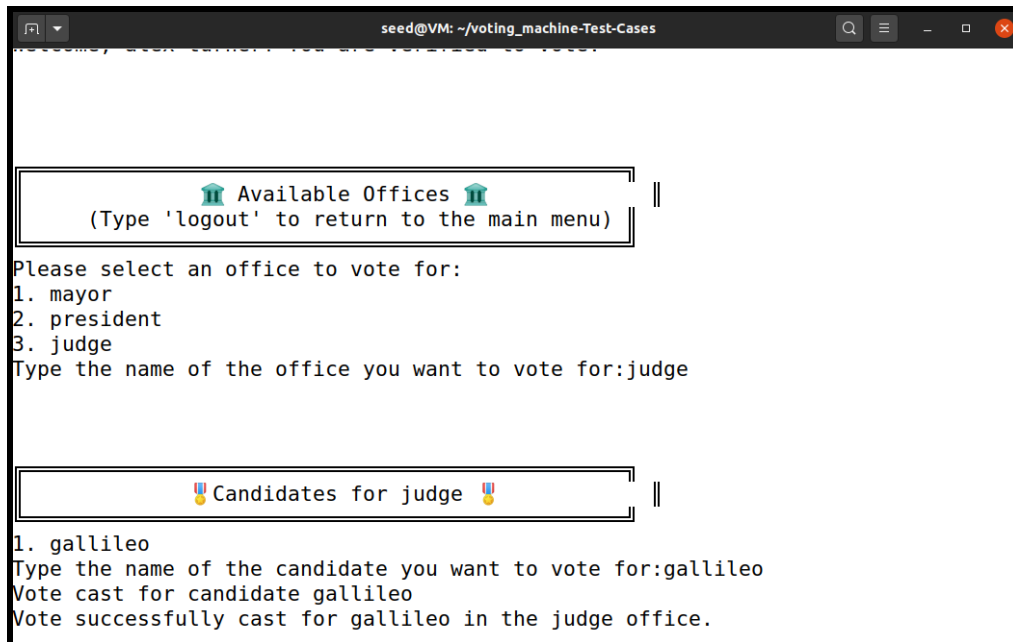


Figure 16: Voting for a candidate- selecting the candidate

11. Type `logout` to exit the system and return to the main menu.

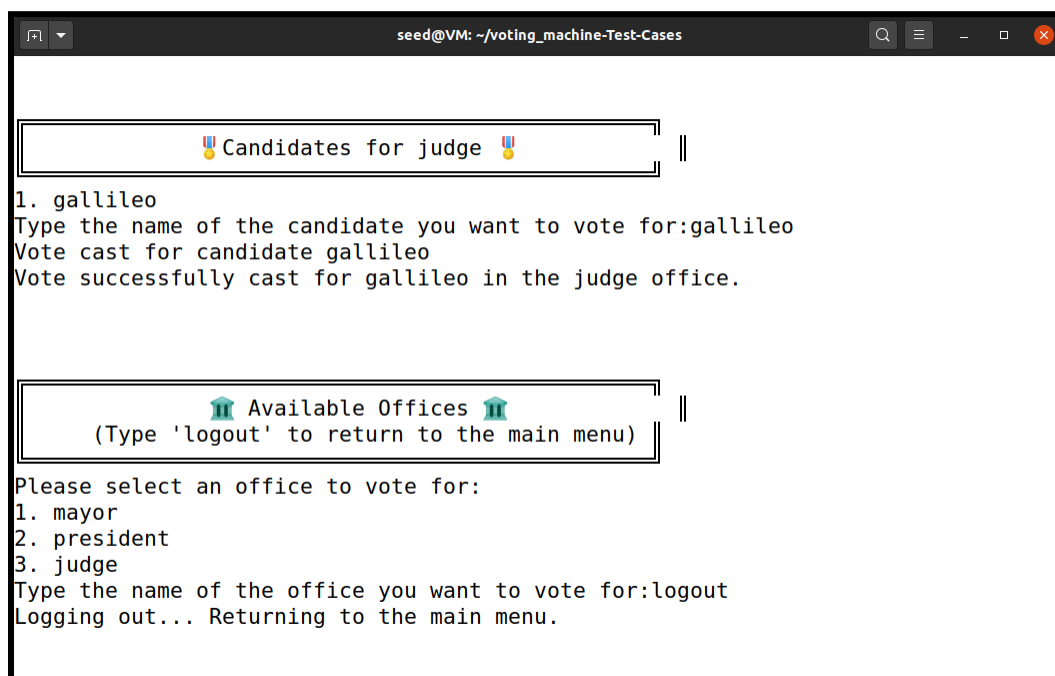


Figure 17: Returning to the main menu

